WEEK 4 · LECTURE L2 · UNIT II

Electronic Recruitment and Applicant Tracking

How the hiring funnel becomes data, whom it quietly screens out, and how to audit it.

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AFTER THIS LECTURE YOU SHOULD BE ABLE TO

What this lecture prepares you for

1. Read a digitized hiring funnel: yield ratios, time to fill and cost per hire, and find where selection really happens.
2. Explain how an ATS works: workflow, matching, candidate relationships, data model and the handoff to the HRIS.
3. Separate speed from validity using Schmidt and Hunter and Sackett et al., and explain three ways an ATS screens people out.
4. Compute and interpret the four fifths rule, distinguish a disparity from its cause, and apply Gilliland’s procedural justice rules.

ONE SOURCE

The same theories as the in class lecture and the course hub, without the simulation data used in class. Course outcomes: CLO 2 and CLO 3.

COMMIT BEFORE THEORY

Commit to an answer first

A vendor tells a Chennai bank that its screening algorithm makes hiring faster and fairer because it removes human judgment. What should the bank ask first?

A

How many applications it can screen in an hour

B

Whether its scores predict job performance, and whether pass rates differ by group

C

Which well known Indian firms already use it

D

Whether its screens work in Telugu and Tamil

EXAM LENS

The answer is B. Speed and language matter, but validity and adverse impact come first.

THE HIRING FUNNEL, DIGITIZED

Attract, apply, screen, shortlist, interview, offer, join

Yield ratio

Candidates passing a stage ÷ candidates entering it. It shows where people drop out.

Time to fill

Days from opening a requisition to the first joiner’s start date.

Cost per hire

Total recruiting cost ÷ number of people hired.

EXAM LENS

Once the funnel is data, every stage can be audited for speed, cost and fairness. The four fifths rule applies at every stage, not only at the offer.

READING A FUNNEL

Where does selection really happen?

Stage	Candidates	Yield from previous stage
Applications	10,000	
Screened	4,000	40%
Shortlisted	1,600	40%
Interviewed	900	56%
Offers	850	94.4% (850 ÷ 900)
Joined	700	82%

WORKED EXAMPLE · COURSE FRAMING

If 94% of interviewed candidates receive an offer, the interview has become a formality: real selection happens at the screen, where only 40% pass. If an algorithm runs there, that is where the audit must start.

INSIDE AN APPLICANT TRACKING SYSTEM

Five parts

1 · Workflow

Requisition approval, posting, screening, assessment, interview scheduling, decision and offer as tracked stages.

2 · Matching

Keywords, skills taxonomies and scores rank candidates, drawn from information retrieval; validity and bias questions follow.

3 · Candidate relationship

Talent pools nurtured over time, borrowed from customer relationship management; every contact shapes the employer brand.

4 · Data model

One requisition, many postings; one candidate, many applications; one application, many interviews. Model it correctly for honest funnel reporting.

5 · Handoff to the HRIS

An accepted offer must create the employee record without rekeying; errors here reach payroll and compliance.

ZOTTOLI AND WANOUS (2000)

Recruitment sources: who enters the funnel

The finding

Hires from inside sources, such as referrals and rehires, have tended to stay longer and perform somewhat better than hires from outside sources.

Realistic information hypothesis

Inside sources give candidates more accurate information about the job, so they choose better and their expectations are met.

Individual differences hypothesis

Different sources reach different kinds of people, so the source shapes who is in the pool at all.

EXAM LENS

Channels decide who can apply before any algorithm runs: a fairness audit that starts at the screen has already missed the first filter. Report time to fill and cost per hire by channel and by location tier.

SCHMIDT AND HUNTER (1998); SACKETT ET AL. (2022)

Speed is not validity

Selection method	Validity (1998)	Revised (2022)
Work sample test	0.54	
Structured interview	0.51	about 0.42
Cognitive ability test	0.51	about 0.31
Unstructured interview	0.38	
Years of job experience	0.18	about 0.07
Years of education	0.10	

EXAM LENS

Validity is the correlation between a selection method and later job performance. A keyword screen rewards degree and years of experience, the two weakest predictors. A fast screen that predicts nothing removes candidates at random, or worse, by group.

THREE WAYS AN ATS SCREENS PEOPLE OUT

Failure modes, and the evidence

Failure mode	How it screens people out	Evidence
Keyword matching	Rejects synonyms, regional job titles and CVs in another language	88% of employers say qualified high skills candidates are screened out for not matching exact criteria (Fuller et al., 2021)
Career gap rules	Filters out returners, often women after caregiving, and seasonal workers	Applicants with family related gaps got fewer callbacks in a field experiment (Weisshaar, 2018)
Poor candidate experience	Long forms that fail on a phone; no status updates; applicants give up	Technology often delivered less than expected (Chapman and Webster, 2003, as reviewed by Bondarouk et al., 2017)

EXAM LENS

Mechanism is not proof: each could cause a gap; only evidence shows that one did. Each can produce adverse impact without any intent to discriminate.

ADVERSE IMPACT

The four fifths rule

	Automated screen	Human screen
Women: offers ÷ applications	100 ÷ 1,000 = 10.0%	140 ÷ 1,000 = 14.0%
Men: offers ÷ applications	300 ÷ 2,000 = 15.0%	300 ÷ 2,000 = 15.0%
Impact ratio	10.0 ÷ 15.0 = 0.67	14.0 ÷ 15.0 = 0.93

EXAM LENS

Impact ratio = lower group’s selection rate ÷ higher group’s selection rate; below 0.80 flags adverse impact (Uniform Guidelines, 1978, section 1607.4(D)). Common errors: comparing offer counts instead of rates, and dividing by all applications instead of each group’s own. Equal rules can still produce unequal rates.

Worked example: hypothetical (Course framing).

A DISPARITY IS NOT YET A CAUSE

From observed gap to conclusion

Observed gap

A ratio of 0.67 under the automated screen, 0.93 under the human screen.

Rival explanations

Different roles, channels or towns in the two groups; a screening rule; the interview stage.

Evidence that decides

Compare like with like (same role and tier); inspect the screening rules; check scores against later performance.

Conclusion

Only then say what caused the gap, and how sure you are.

EXAM LENS

The Uniform Guidelines also warn that small samples can mislead in both directions.

GILLILAND (1993)

Candidate experience is a fairness question too

Job relatedness

Does the screen visibly measure what the job needs?

Opportunity to perform

Can the applicant show what she can do, or is she judged on keywords alone?

Feedback

Does she learn the outcome, and when?

Two way communication

Can she ask a question or contest a decision?

EXAM LENS

Applicants judge fairness by the process as well as the outcome. These are four of Gilliland’s ten procedural rules for selection systems; a silent rejection breaks several at once.

BEFORE YOU BUY

Auditing a vendor’s claims

1

What does the score predict, and against which measure of job performance?

2

What validation evidence exists for people like our applicants?

3

What are the selection rates by group at each stage, not only at the offer?

4

What data trained the model, and where will our applicants’ data be stored?

5

Can a person override the score and explain a rejection?

Why ask

Raghavan et al. (2020): vendors disclose little about validation or bias testing, so the buyer must ask.

THEORY IN DEPTH · GRIGGS V. DUKE POWER CO. (1971)

Results count, not intentions

The core idea

The US Supreme Court held that neutral looking tests are unlawful if they exclude groups and are not job related.

How it works

This disparate impact doctrine led to the 1978 Uniform Guidelines and the four fifths rule.

Limits and critique

The four fifths rule is a rule of thumb, not a statistical test; small samples can mislead.

IN PRACTICE

An algorithm need not intend bias; what matters is whether its outcomes differ by group without job related justification.

EXAM LENS

This is US law; Indian law has no direct equivalent, but the analytical logic of disparate impact still applies.

THEORY IN DEPTH · WANOUS (1973); PREMACK AND WANOUS (1985)

Telling the truth about a job improves retention

The core idea

A realistic job preview shows both the good and the hard parts of a job before the offer.

How it works

Candidates self select out early, and those who join have accurate expectations met.

Limits and critique

The effect on turnover is real but modest in meta analysis.

IN PRACTICE

For high attrition frontline roles, a short phone video of a real shift may do more than another screening filter.

EXAM LENS

Report the meta analytic effect honestly: modest, but reliable.

A hiring model can learn past bias

The core idea

Amazon dropped an experimental resume scoring tool after it was found to downgrade resumes mentioning women’s activities.

How it works

Trained on a decade of mostly male hires, it learned to treat male linked signals as merit.

Limits and critique

Removing gender words did not fix it, because other features acted as proxies.

IN PRACTICE

Any vendor ATS was trained on someone’s past data; ask whose, and audit its outcomes by group.

EXAM LENS

Use the case to illustrate proxy discrimination, then link it to Barocas and Selbst’s routes (Week 5).

A WARNING

The same pass rate can hide different fairness

The trap

Two groups of requisitions, with and without an algorithm, can show almost identical overall screening pass rates.

Where the gap appears

Only when offers are split by gender and selection rates compared does an adverse impact become visible.

The test

Analyze each stage with rates split by group, not totals: applications (channel reach), screening (the most automated stage), shortlist and interview, offer and joining (acceptance and early attrition).

EXAM LENS

A screen can be fast, valid and still unfair: validity and fairness are different questions; test both.

APPLY

An ATS for 50,000 warehouse applicants

Teaching case (Course framing)

A Hyderabad quick commerce firm hires 3,000 warehouse associates a year from about 50,000 applicants. A vendor ATS scores English CVs on keywords such as forklift, inventory and WMS, and rejects applicants with a gap of more than six months. Recruiters love its speed. Applicants apply mostly by phone, many in Telugu or Hindi.

Work it through in pairs

1. Whom will this screen reject before a person reads the CV?
2. Which predictors does a keyword and gap rule rely on, and how valid are they?
3. How would you test for adverse impact, and at which stage?
4. Which of Gilliland’s procedural rules does a silent rejection break?

COMMON CONFUSIONS

Tempting readings, and better ones

A tempting reading	A better reading
A fast screen is a good screen	A screen is useful only if it predicts job performance
Equal numbers of offers mean the process is fair	Compare selection rates, each divided by that group’s own applications
A ratio below 0.80 proves discrimination	It flags adverse impact that must be explained; the cause needs evidence
Fairness starts at the screen	Channels decide who applies before any screen runs

COMPARE AND EVALUATE

Six lenses on digital hiring

Lens	Question it answers	Watch out for
Funnel yield ratios	Where does selection happen?	Stage definitions must be consistent *
Zottoli and Wanous (2000)	Who enters the funnel?	Effects are modest
Schmidt and Hunter; Sackett et al.	Does the method predict performance?	Estimates revised downward in 2022
Four fifths rule (1978)	Is there adverse impact?	Flags, does not prove; small samples mislead
Gilliland (1993)	Does the process feel fair?	Perceptions, not validity
Raghavan et al. (2020)	What should vendors disclose?	Disclosure is voluntary

* Course framing.

PRACTICE FOR THE END TERM EXAMINATION

Questions to rehearse

A Bengaluru retailer reports that its new ATS cut time to hire by a week. Using selection validity, the four fifths rule and applicant reactions, evaluate whether the ATS improved hiring, and state what evidence you would still need. **20 marks**

A strong answer: separates speed from validity and explains why CV proxies predict weakly; computes selection rates from each group's own applications and applies 0.80 correctly; uses Gilliland for phone only and vernacular applicants; separates disparity from cause.

Explain how an applicant tracking system can produce adverse impact without any intent to discriminate, and design an audit. **20 marks**

A strong answer: names keyword, gap and experience failure modes; audits every stage and channel; computes impact ratios; names the evidence that would establish cause.

In a funnel, 900 candidates are interviewed and 850 receive offers. What does this tell you about where selection happens? **10 marks**

A strong answer: computes 94.4%; concludes the interview is a formality and selection happens at the screen, where an audit should start.

CARRY THESE INTO THE EXAMINATION

Four things to remember

1. **Find where selection really happens.**
   Yield ratios show the stage that decides.

2. **Speed is not validity.**
   CV proxies predict weakly; ask what the score predicts.

3. **Rates, by group, at every stage.**
   Below 0.80 flags adverse impact; it does not prove cause.

4. **The process is part of fairness.**
   Channels, language, phones and feedback decide who is heard.

SET READINGS · FROM THE LESSON PLAN

Read for this week

- Bondarouk, T., Parry, E., & Furtmueller, E. (2017). *IJHRM, 28*(1), 98 to 131.
- Kavanagh, M. J., & Johnson, R. D. (2018), Chapters 1 and 3.
- An Indian HRIS or ATS implementation case.

NOTE

The other works named in this lecture are theory sources for reference, not set readings.

THEORY SOURCES · FOR REFERENCE

Sources for this lecture

- Dastin, J. (2018, October 10). Amazon scraps secret AI recruiting tool that showed bias against women. Reuters.
- Bondarouk, T., Parry, E., & Furtmueller, E. (2017). Electronic HRM: Four decades of research on adoption and consequences. *The International Journal of Human Resource Management, 28*(1), 98 to 131. https://doi.org/10.1080/09585192.2016.1245672
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- Fuller, J. B., Raman, M., Sage Gavin, E., & Hines, K. (2021). *Hidden workers: Untapped talent*. Harvard Business School Project on Managing the Future of Work and Accenture.
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- Gilliland, S. W. (1993). The perceived fairness of selection systems: An organizational justice perspective. *Academy of Management Review, 18*(4), 694 to 734. https://doi.org/10.2307/258595
- Raghavan, M., Barocas, S., Kleinberg, J., & Levy, K. (2020). Mitigating bias in algorithmic hiring: Evaluating claims and practices. *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*, 469 to 481. https://doi.org/10.1145/3351095.3372828

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- Weisshaar, K. (2018). From opt out to blocked out: The challenges for labor market reentry after family related employment lapses. *American Sociological Review, 83*(1), 34 to 60. https://doi.org/10.1177/0003122417752355
- Wanous, J. P. (1973). Effects of a realistic job preview on job acceptance, job attitudes, and job survival. *Journal of Applied Psychology, 58*(3), 327 to 332. https://doi.org/10.1037/h0036305
- Zottoli, M. A., & Wanous, J. P. (2000). Recruitment source research: Current status and future directions. *Human Resource Management Review, 10*(4), 353 to 382. https://doi.org/10.1016/S1053-4822(00)00032-2